

PARAMETRIC AND NON-PARAMETRIC ANALYSIS

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Introduction

- Statistical Analysis is a universal method with which we can assess the validity of a conclusion. It is one of the most important aspects of a scientific research.
- The statistical methods used in quantitative research can be described as either **parametric** or **nonparametric**.
- Both parametric and non parametric analysis have their purposes, advantages and limitations.

What is Parametric Analysis ?

- Parametric analysis is a branch of inferential statistics wherein one obtains a sample from a population in order to estimate population parameters (e.g., mean) and investigate relationships between the estimated parameters.
- Because this estimation process involves a sample, a sampling distribution, and a population, certain parametric assumptions are required to ensure all components are compatible with each other.

Different Parametric Tests

- **1. T-Test - Assumptions of this test:**
 - Population distribution is normal,
 - Samples are random and independent
 - Sample size is small.
 - Population standard deviation is not known.
- **2. Z-Test - Assumptions of this test:**
 - Independent observations
 - Interval or ratio level data
 - Normal population or large N (at least 30)

Different Parametric Tests

● 3. F-Test - Assumptions of this test-

- The populations from which the samples were obtained must be normally or approximately normally distributed.
- The samples must be independent of each other.
- The variances of the populations must be equal.

● 4. ANOVA – Assumptions of the test-

- Independence of observations.
- The variances of the populations must be equal.
- The dependent variable is normally distributed in each group that is being compared in the one way ANOVA test.

Advantages of Parametric tests

- Provides accuracy and precision to the results.
- Performance based on distribution.
- There is complete information about the population.
- It has highest statistical power as compared to other tests.

Disadvantages of parametric tests

- There is influence of sample size,.
- Parametric tests are not valid when it comes to small no of samples (if $n \leq 10$).
- Parametric test is used only for interval data and ratio data.
- Requirement of mathematical calculation is high.
- Susceptible to outlier.

What is Non Parametric Analysis ?

Non parametric analysis or distribution free tests are those that do not rely on parameter estimates or precise assumptions about the distribution of variables. Non parametric Analysis can be used for descriptive statistics or inferential statistics.

Non Parametric Tests

- **Mann Whitney U test**

- For randomly selected values X and Y from two populations, the probability of X being greater than Y is equal to the probability of Y being greater than X .

- **Kruskal-Wallis test**

- For testing whether samples originate from the same distribution. It is used for comparing two or more independent samples of equal or different sample sizes.

- **Sign test**

- to test for consistent differences between pairs of observations, such as the weight of subjects before and after treatment.

- **Wilcoxon Signed-Rank test**

- to test the **location** of a population based on a sample of data, or to compare the locations of two populations using two matched samples.

Non Parametric Tests

- **Assumptions of these tests are similar –**
 - The sample drawn from the population is random.
 - The independence within the samples and mutual independence is assumed. Ordinal measurement scale is assumed.

Advantages of Non Parametric tests

- Non parametric tests are not susceptible to outliers –like parametric test.
- They do not require restrictive assumptions.
- They are simple and easier to calculate.

Disadvantages of parametric tests

- Non parametric tests are less powerful when the data follows **Gaussian** distribution.
- As the sample size gets **larger** data manipulations required for non parametric tests becomes **laborious**.
- Appropriate computer software for nonparametric methods are not too many, although the situation is improving.

Parametric vs non Parametric: Comparison

Non-parametric

- Can be used on ordinal and nominal scale data (although also on interval and ratio scale).
- Can be used on small samples.
- Can be used on data that are not normally distributed.
- Can be used where the samples are not selected randomly.
- Have less power than the equivalent parametric test.

Parametric

- Used mainly on interval and ratio scale data.
- Tend to need larger samples.
- Data should fit a particular distribution; the data can be transformed to that distribution.
- Samples should be drawn randomly from the population.
- More powerful than non-parametric equivalent.

References

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THANK YOU